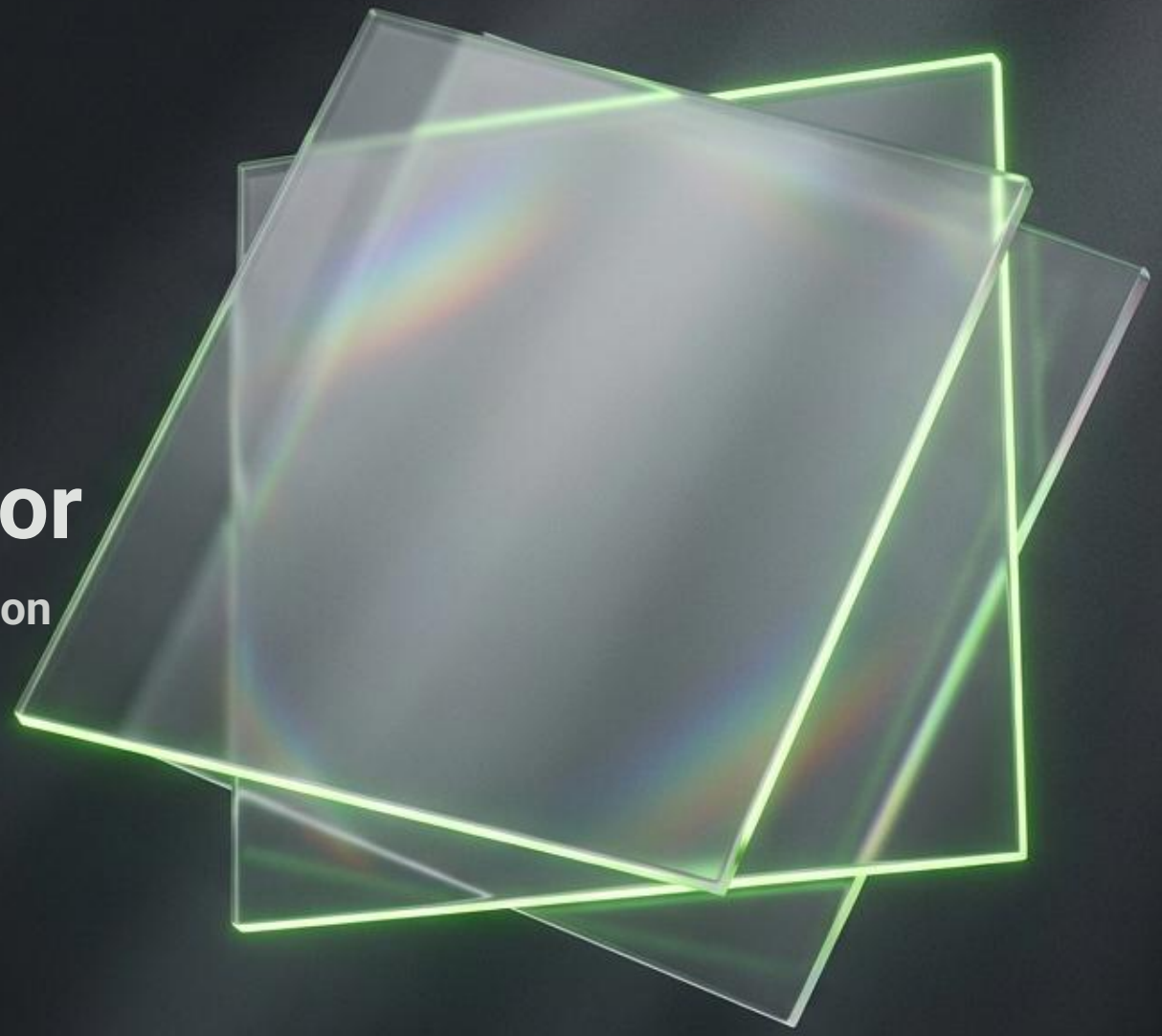
Synthetic Fraud Generator

Balance datasets safely to improve fraud detection



Presented by

FraudX

Agenda

Navigating the story of data-driven security

01. The Challenge

Why fraud data scarcity and imbalance hurts detection accuracy.

02. The Solution

Generating synthetic fraud patterns with privacy preserved.

03. Workflow & Use Cases

End-to-end process from data loading to model retraining.

04. Impact & Audience

Benchmarks for bank risk teams and fraud analysts.

The Fraud Data Problem

With the context set, we define the core problem

⚠️ Severe Imbalance

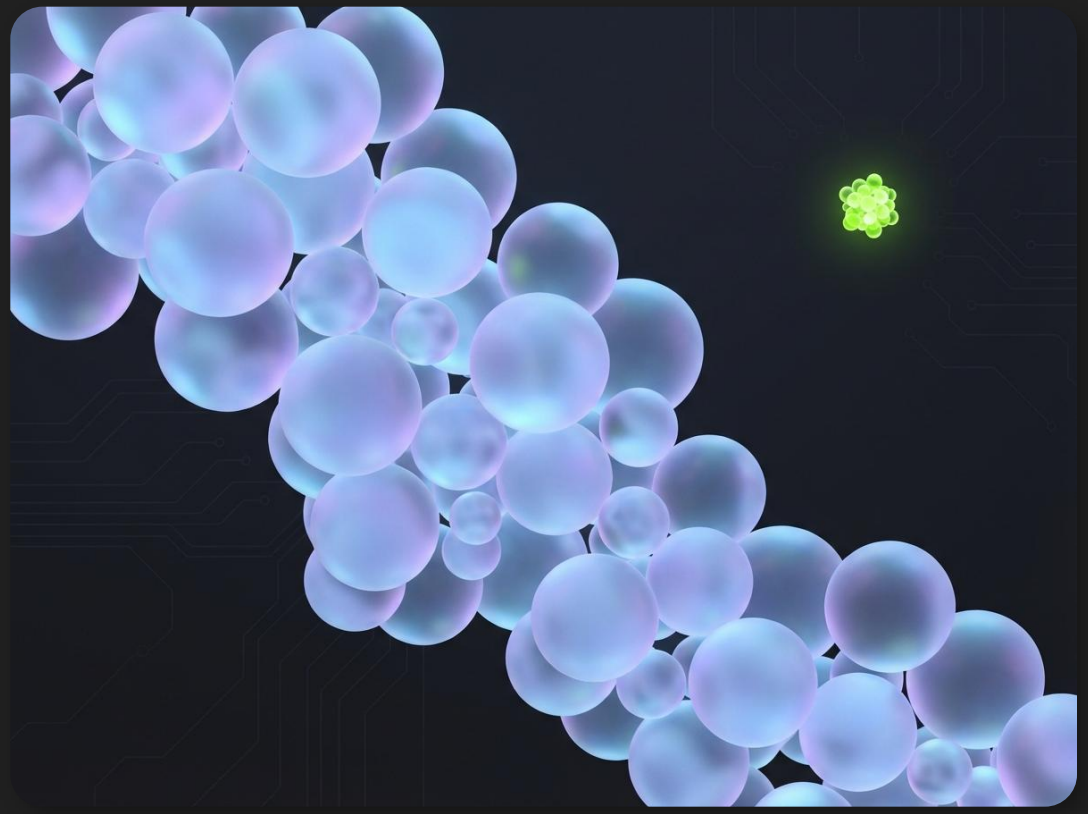
Financial datasets are often overwhelmingly normal, with fraud cases making up less than 1% of transactions.

📉 Under-Detection

Models trained on such skewed data struggle to learn fraud patterns, leading to missed detections.

🔒 Privacy Restrictions

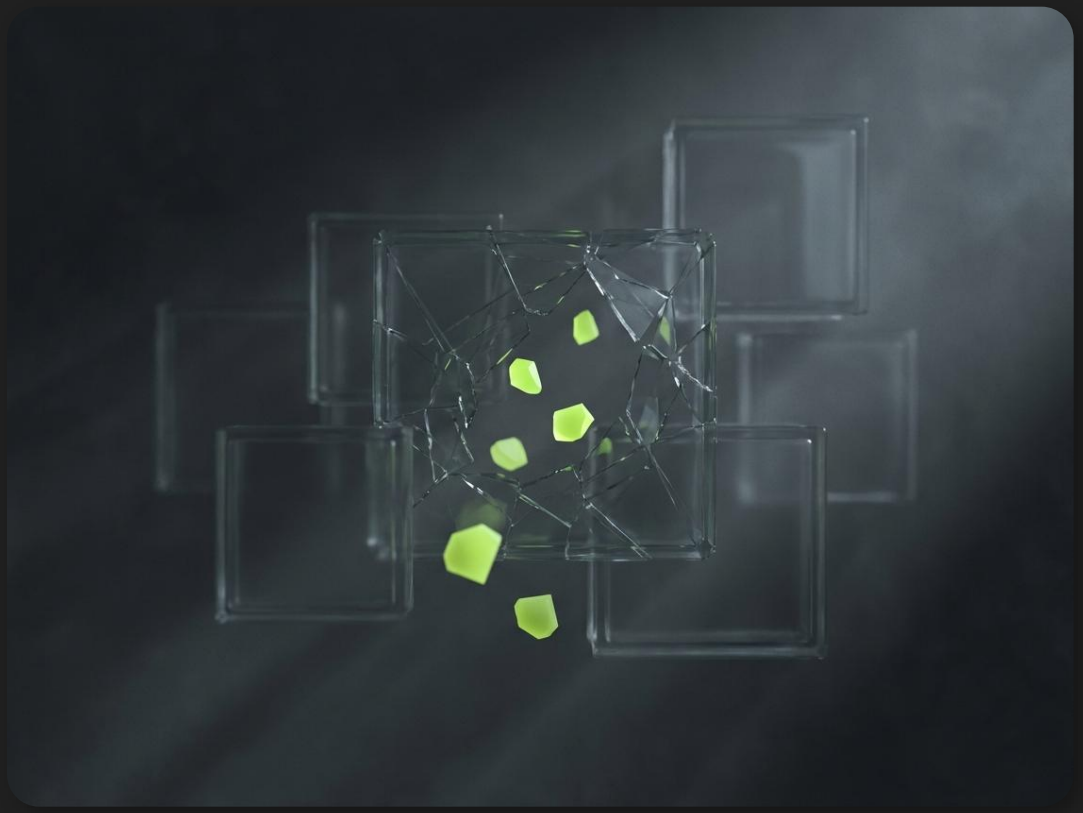
Real fraud data is sensitive and heavily restricted, limiting the ability to share or augment datasets easily.



Visualizing the scarcity of fraud signals (green)

Why It’s Critical

The consequences of ignoring data imbalance



🔍 High False Negatives

Models learn to favor the majority class (normal transactions), failing to flag actual fraudulent activities.

💸 Financial & Reputational Loss

Every missed fraud case results in direct financial loss and damages the institution's reputation.

🤖 Erosion of Trust

Consistent failures in detection erode trust in AI-enabled risk management systems.

Synthetic Fraud Generator

Our approach to bridging the data gap

✨ AI-Powered Generation

We use algorithms to generate artificial yet statistically realistic fraud transactions to balance the dataset.

🛡️ Privacy-First Training

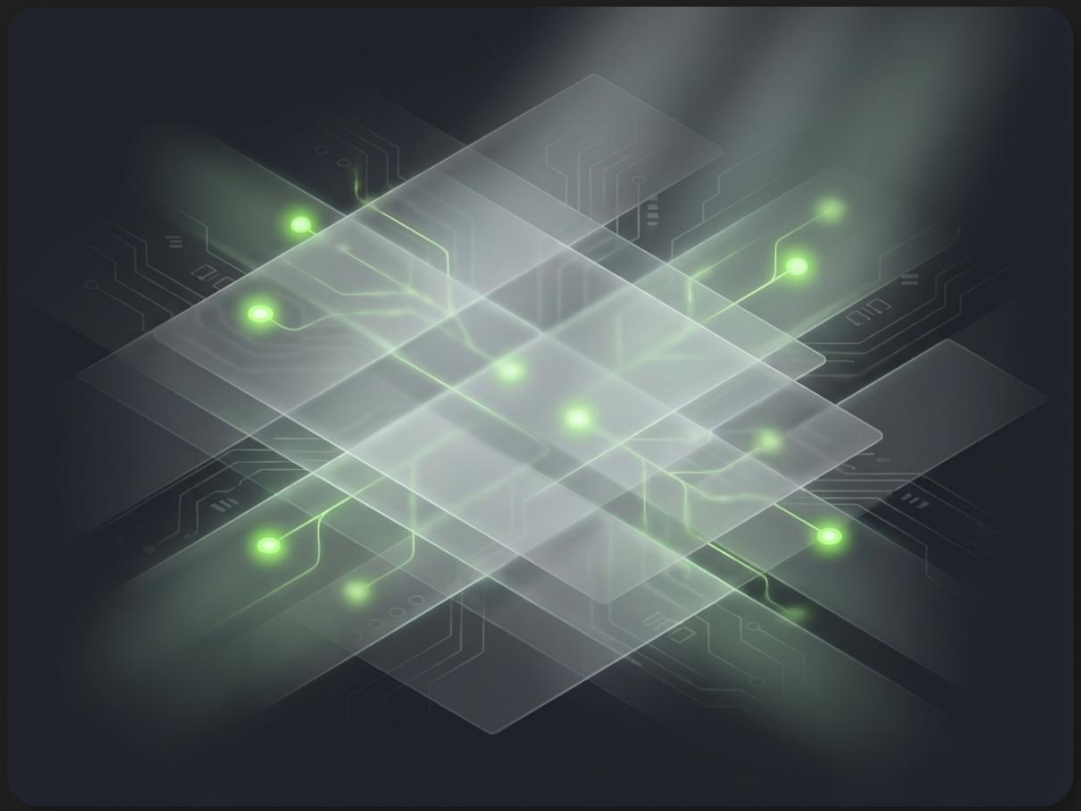
Synthetic data is used purely for training the model, ensuring no real user data is exposed or compromised.

📈 Enhanced Learning

By balancing the classes, the model can learn robust decision boundaries without being biased towards the majority class.

Understanding Synthetic Data

What makes it safe and effective?



🧩 Mimicry, Not Copying

Synthetic data replicates the statistical properties and correlations of real fraud without duplicating specific user records.

🔐 Pattern-Based Safety

Generative processes learn abstract patterns (e.g., high-value transaction at odd hours) rather than memorizing identities.

🚀 Better Generalization

This helps models generalize to new, unseen fraud attempts rather than overfitting to a few known historical cases.

How FraudX Works

From raw data to robust detection

1

Load Data

Ingest transaction logs

2

Identify

Isolate minority class

3

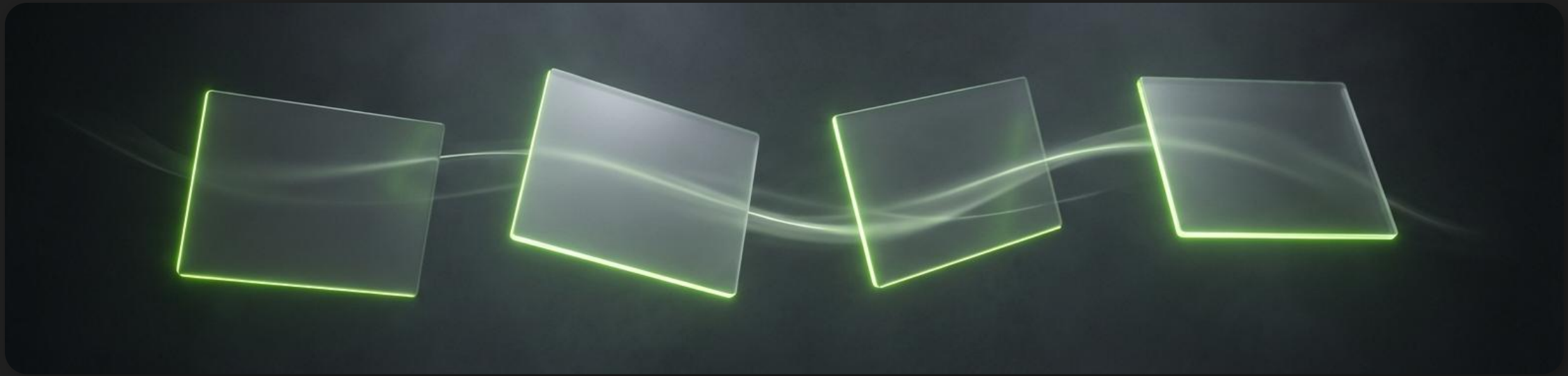
Generate

Create synthetic samples

4

Retrain

Update model weights



Transaction Dataset

A look at the data powering our demo

Source

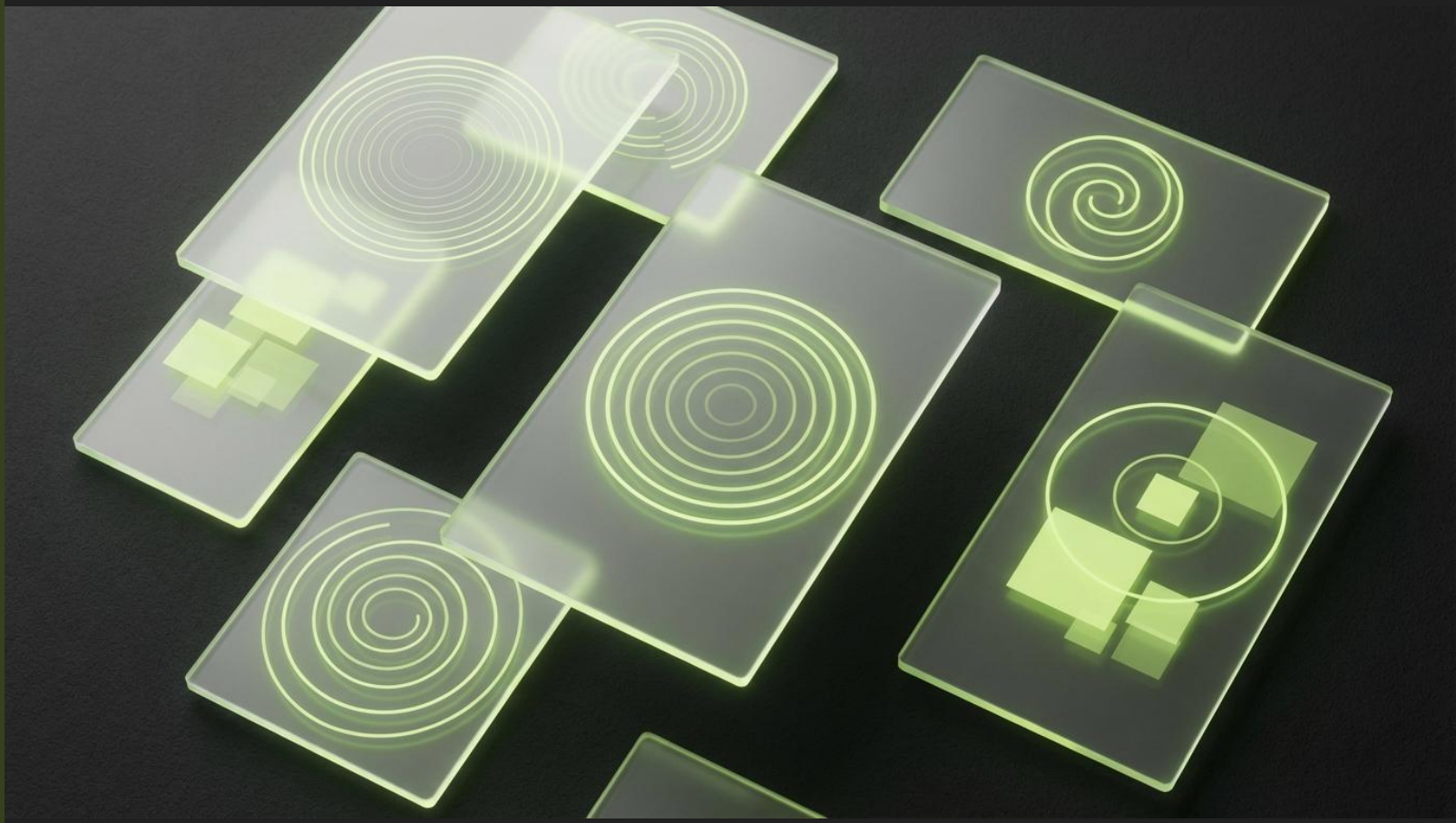
Public credit card transaction repository (e.g., Kaggle).

Key Features

- Transaction Amount
- Timestamp
- 28 Anonymized V-features

Target

Binary Classification (0: Legit, 1: Fraud)



Tech Stack Used

Synthetic Minority Over-sampling Technique

🔗 **Language**

Python — core development language

🔗 **AI / Machine Learning**

Scikit-learn — fraud detection model training

🔗 **Data Processing & Visualization**

Pandas, NumPy — data handling and preprocessing

Matplotlib / Seaborn — performance analysis and visualization

🔗 **Development Environment**

Google Colab — cloud-based execution

SMOTE Technique

Synthetic Minority Over-sampling Technique

🔗 Interpolation Logic

Creates new fraud instances by drawing lines between existing fraud points and picking a spot along the line.

⚡ Lightweight & Fast

Computationally efficient compared to GANs or VAEs, making it perfect for rapid iteration.

📏 Low Complexity

Does not require massive GPU resources or deep learning infrastructure to deploy.

Training Flow

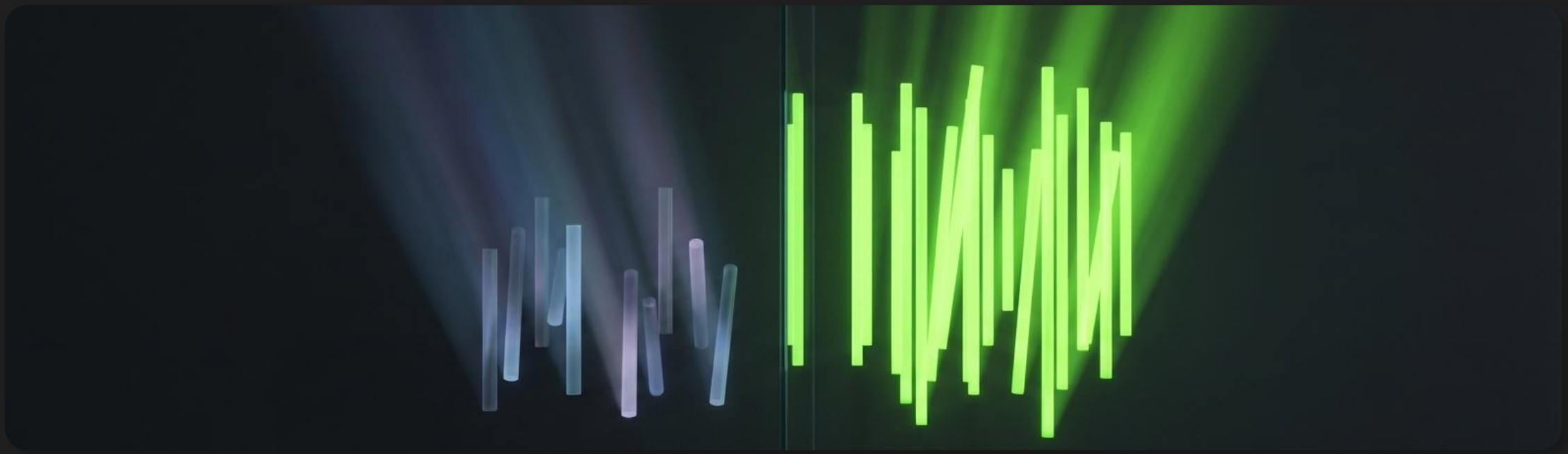
Baseline vs. Augmented Performance

Before (Baseline)

Model trained on raw, imbalanced data. Struggles to find the signal in the noise.

After (Augmented)

Model trained with synthetic fraud samples. Clearer decision boundaries and higher recall.



Results & Benchmarks

Measuring the impact

🚀 Increased Recall

Fraud detection rate improved significantly (e.g., from 60% to 85%), catching more bad actors.

📉 Fewer False Negatives

Drastic reduction in missed fraud cases, directly saving potential financial losses.

⚖️ Stability

Performance remains consistent across different time windows, proving the model isn't just memorizing.

Conclusion

Ready to scale for safer fintech

Privacy-preserving, data-balancing, and effective.